Lab brief: does force speed the change?

Same total mass. Different driving force. F = ma emerges.

12 A

TASK 1 · KEEP THE MASS FIXED

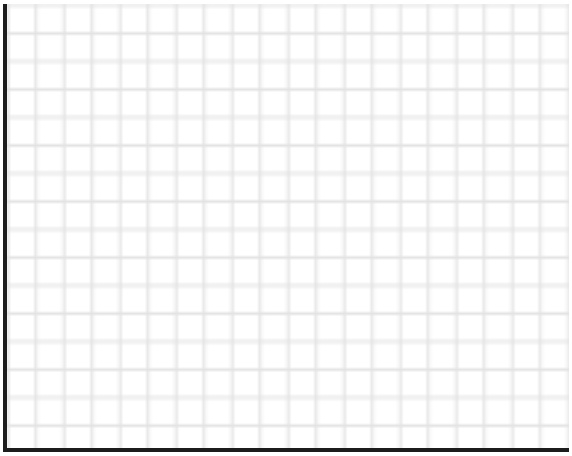
Move masses from the cart to the hanger. Never add new ones. Total moving mass: ____ kg

- ☐ total mass unchanged
- ☐ release from rest
- ☐ track level
- ☐ 2+ trials each

TASK 2 · MY DATA

HANGING MASS (kg)	DRIVING FORCE (N)	TRIAL 1 a	TRIAL 2 a	AVERAGE a

TASK 3 · GRAPH: a vs. FORCE



TASK 4 · READ THE SLOPE

Slope = ____ (m/s²)/N. Mass = 1 ÷ slope = ____ kg. Does it match your total?

ANALYST SIGN-OFF · COMBINE BOTH LABS

a ∝ F and a ∝ 1/m, so a = ____ ÷ ____. My evidence: ____

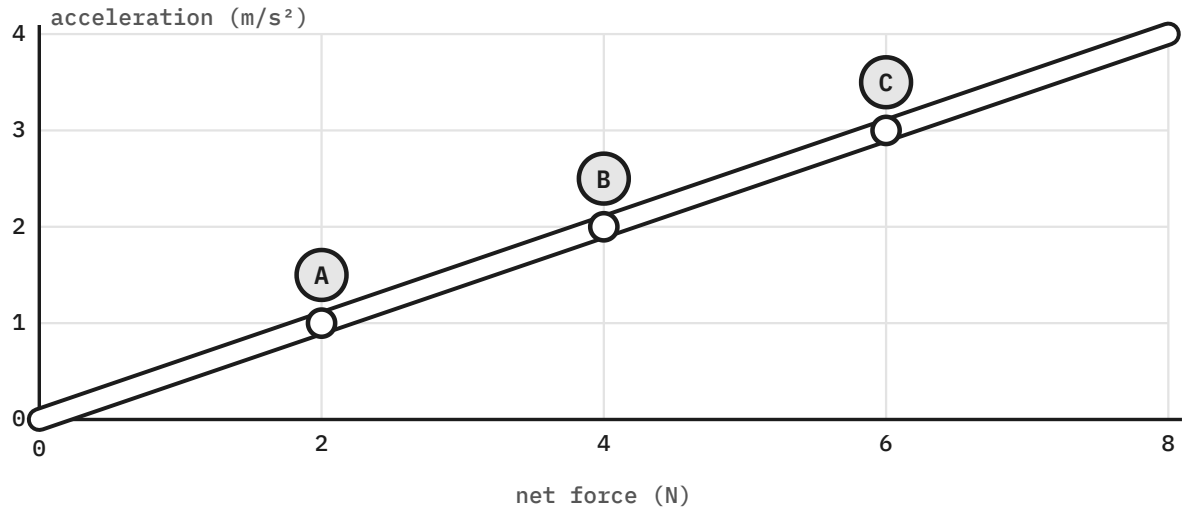
S4 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Read the a vs. F graph

12 F

DIRECTIONS · Ideal 2 kg system (illustrative). Real data may sit lower because of friction.



A · 2 N

a = ____ m/s²

B · 4 N

a = ____ m/s²

C · 6 N

a = ____ m/s²

SLOPE

____ (m/s²)/N → m = ____ kg

PREDICT

Same system, 10 N. a = ____ m/s². What if mass doubled to 4 kg at 10 N? a = ____

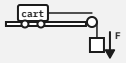
WHY MIGHT REAL DATA MISS THE LINE?

S4 1 2 3

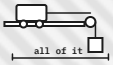
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Say it, then write it: the force lab

Language goal: I can **compare** two results.



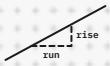
driving force
fuerza impulsora · força motriz
the hanging weight that pulls



total mass
masa total · massa total
cart + hanger + all masses



proportional
proporcional · proporcional
double one, double the other



slope
pendiente · inclinação
rise ÷ run of a graph line

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: force · mass · doubles · proportional · slope

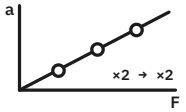
We changed the driving ____ but kept the total ____ the same. When the force doubles, the acceleration ____. So a and F are ____. The ____ of the a vs. F line is 1 ÷ m.

2 · SAY IT TO A PARTNER, THEN WRITE IT



When the force was ____ N, the acceleration was ____ m/s².
When the force doubled, a ____, so they are ____ .

3 · SOLVE WITH THE STEPS



Same 2 kg system: 2 N gives 1 m/s². Predict a at 6 N.

STEP 1 · COMPARE
6 N ÷ 2 N = × ____

STEP 2 · SCALE a
1 × ____ = ____ m/s²

STEP 3 · SAY IT
"Three times the force gives ____ the acceleration."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ Acceleration is proportional to force because ____ .
☐ We moved masses to the hanger so that ____ .
☐ My data point at ____ N was off because ____ .

I said it ☐ · I wrote it ☐

S4 1 2 3

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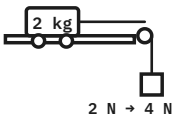
UNIT 1 · DAY 12 · S4 · SEI · PARTNER TALK

Talk it through: the force lab

Language goal: I can **compare** two results.

DIRECTIONS · Read your lines aloud. Listen, then write. Switch roles each round.

ROUND 1 · 2 N → 1 m/s², 4 N → ? (SAME MASS)



A ASKS "What happens to a when the force doubles?"

B SAYS "The force goes from ____ to ____, so a goes from 1 to ____ m/s²."

A CHECKS "So you're saying a and F are ____."

ROUND 2 · SWITCH · WHAT DID WE KEEP THE SAME?

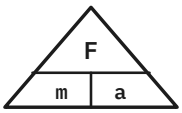


B ASKS "Why did we move masses instead of adding them?"

A SAYS "To keep the ____ the same. If we add mass, ____ changes too."

B ADDS "I want to add that the hanger's ____ is the driving force."

ROUND 3 · BOTH · PUT BOTH LABS TOGETHER



A "More mass gives ____ acceleration."

B "More force gives ____ acceleration. So a = ____ ÷ ____."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

F = ma means…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

S4 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Pitch "PushBot"

Your warehouse robot pushes carts. Investors want to know how fast it can speed them up.

12 Z

SLIDE 1 · TEST DATA (robot pushes with a steady net force)

CART	MASS (kg)	NET FORCE (N)	a = F ÷ m
empty	20	60	
half full	40	60	
full	60	60	
full + boost	60	120	

SLIDE 2 · THE PATTERN

Doubling the mass ____ a. Doubling the force ____ a.

SLIDE 3 · SPEC

A full cart must reach 2 m/s in 2 s. Needed a = ____ . Needed F = ____.

THE PITCH · 30 SECONDS

"PushBot uses F = ma: a heavier cart needs ____ force for the same acceleration. For a full cart we deliver ____ N, so it reaches ____ m/s in ____ s."

SHARK QUESTION

"Why not just push harder on every cart?" Your answer:

S4 1 2 3

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Variable, control, or result?

12 N

DIRECTIONS · Today's lab: change the driving force, keep the total mass. Sort each card.

A hanging mass (driving force)	B total moving mass	C acceleration	D track angle (level)
E velocity–time slope	F release method	G number of masses on the hanger	H start position
I the a vs. F graph	J cart + hanger + all masses	K friction in the setup (try to keep it)	L average a of trials

CHANGED	CONTROLLED	MEASURED
what we vary	kept the same	the result

EXPLAIN ONE TRICKY CARD

Card ____ belongs where it does because _____

S4 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Comic: a bigger push, a bigger change

12 CS

YOUR READER

A younger student who thinks a bigger push makes a cart go fast instantly.

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◌ **ORANGE** — Force arrows; the panel 2 arrow is twice as long
- ☐ ◌ **GREEN** — Both carts, because both speed up
- ☐ ◯ **PENCIL** — Graph axes labeled **F (N)** and **a (m/s²)**

1 · SMALL PUSH *A cart pushed with a small force.*

CAPTION _____

2 · BIGGER PUSH *The same cart pushed twice as hard.*

CAPTION _____

3 · THE GRAPH *A character shows the a vs. F graph.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **force · acceleration · proportional · doubles**

S4 1 2 3

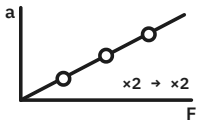
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Doodle Notes · proportional relationship

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Proportional relationship · relación proporcional (ES) · relação proporcional (PT)



TODAY'S CASE

A 2 kg cart experiences net forces of 2 N, 4 N and 6 N in three trials.

01 · SKETCH AND LABEL

Sketch the trials with orange arrows to scale. Label what stays fixed.

02 · CHANGE THE PICTURE

Plot the three force–acceleration points. Name both axes and use the origin to show what doubling the net force does.

Annotate your sketches. Circle the mark that best shows proportional relationship. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

When net force doubles at fixed mass, acceleration …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Frayer Model · proportional relationship

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

Proportional relationship · relación proporcional (ES) · relação proporcional (PT)

A · At fixed mass, twice the net force gives twice the acceleration.

B · Any two quantities that both increase must be directly proportional.

01 · DEFINITION

What does proportional relationship mean in your own words? Use the notes or start in your home language.

02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

SAY IT TO A PARTNER · THEN WRITE

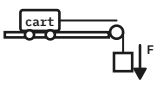
I chose card ___ as my example because …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
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Picture Dictionary · proportional relationship

I can connect each word to a picture and use it to explain this case.

Name: _____ My home language: _____

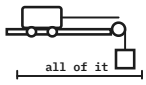


driving force

fuerza impulsora · força motriz

the hanging weight that pulls

My word or label: _____

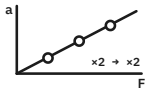


total mass

masa total · massa total

cart + hanger + all masses

My word or label: _____

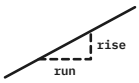


proportional

proporcional · proporcional

double one, double the other

My word or label: _____



slope

pendiente · inclinação

rise ÷ run of a graph line

My word or label: _____



acceleration

aceleración · aceleração

change in velocity per second

My word or label: _____



friction

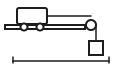
fricción · atrito

contact force that opposes slipping or a tendency to slip

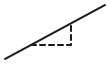
My word or label: _____

MATCH · WRITE THE ENGLISH WORD UNDER EACH PICTURE

_____



_____



_____

Use two words to explain the case: A 2 kg cart experiences net forces of 2 N, 4 N and 6 N in three trials.

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else